Phase 2 - Week 5 - Lesson Day 2

Linear Regression

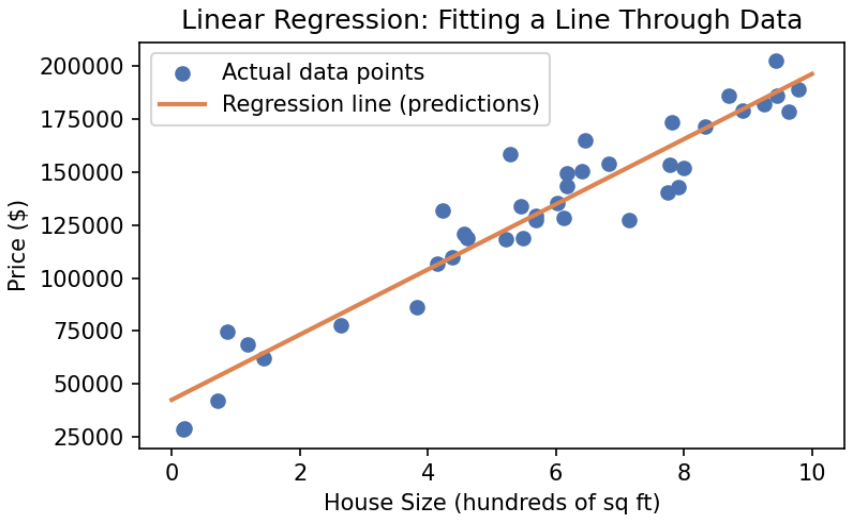
Today's Goal

Today you build your first real machine learning model. **Linear Regression** predicts a number (like a price or a score) based on one or more input values.

- Understand what linear regression does
- Train a linear regression model with Scikit-learn
- Use the model to make predictions

1. What Is Linear Regression?

Simple definition: Linear regression finds the best straight line that fits through your data points. Once it has that line, it can predict a number for any new input - even inputs it has never seen before.



In the picture above, the blue dots are real house sizes and prices. The orange line is what the model learned - the "best fit" line through the data. To predict a new house's price, the model just finds where that house's size lands on the line.

2. Training a Linear Regression Model

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

# X = input (house size), y = target (price)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42
)
```

```
model = LinearRegression()
model.fit(X_train, y_train)   # this is the 'training' step - the model learns the best
line
```

.fit() is the key step in almost every Scikit-learn model. It means "learn the pattern from this training data." You will see **.fit()** used the same way for every model you build in this course.

3. Making Predictions

```
predictions = model.predict(X_test)   # predict prices for the TEST houses (unseen by th
e model)

print(predictions[:5])   # show the first 5 predicted prices
print(y_test[:5])         # compare to the actual prices
```

.predict() is the second key step - you give it new inputs, and it gives back predicted outputs, using the pattern it learned during **.fit()**.

You can also check how good the model's predictions are using **.score()**, which returns a number (called R-squared) between 0 and 1 - closer to 1 means the model explains the data very well.

```
print(model.score(X_test, y_test))   # closer to 1 = better fit
```

Today's Practice

Open **day-2-linear-regression.ipynb**. All answers are already solved.

Practice Questions (5 total - all solved)

1. Build a dataset

Create a made-up dataset: years_experience -> salary (similar to Day 1).

2. Split and train

Split the data into train/test, then train a LinearRegression model on the training data.

3. Predict

Use the trained model to predict salaries for the test data, and print the first 5 predictions next to the first 5 actual values.

4. Score the model

Check the model's score using .score() and print it.

5. Predict a brand new value

Predict the salary for someone with 7 years of experience, a value that wasn't directly in your dataset.

Once you and Halimo finish reviewing today's practice, move to Lesson Day 3 (Logistic Regression) - predicting categories instead of numbers.